COUNTEREXAMPLES, SPECTRAL OBSTRUCTIONS, AND DELETION STABILITY FOR WOW-284

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ABSTRACT. WOW-284 predicts that the minimum dual degree of every connected graph of order at least three and girth at least five does not exceed the negative of its least distance eigenvalue. We give exact counterexamples of orders $38, 39, 40, 42$, and 50, beginning with the Hoffman–Singleton graph, and develop a structural theory of the failure. For a regular graph of girth at least five and diameter three, the counterexample score is

$$\delta^*(G) + \lambda_{\min}(D(G)) = 2k - 2 - \max_{\theta \neq k}(\theta + 1)^2,$$

where θ ranges over the nonprincipal adjacency eigenvalues. Every regular strict counterexample has degree at least six and diameter at most four; diameter-four counterexamples are excluded through degree nine. We determine the exact optimum, and the unique optimizer up to scale, of the standard one-variable nonbacktracking linear-programming bound. Localizing its extremal polynomial at an edge yields a five-cycle certificate which excludes degree-six order 51, and hence every degree-six regular strict counterexample has order at most 50. We further derive complete distance spectra for one- and two-vertex punctures of Moore graphs, prove a general deletion-stability inequality, and give a metric normal form for arbitrary punctures of size at most $k-1$. In particular, every deletion of at most five vertices from the Hoffman–Singleton graph remains a strict counterexample, while one explicit six-vertex deletion does not. Equality cases, finite-field obstructions, matching-deletion controls, and exact local constraints on a hypothetical degree-six order-50 example are also obtained. All theorem-level computations use exact arithmetic; independent Python certificates are cited inline, and the explicit finite counterexamples have Lean 4.31 kernel-checked certificates.

1. INTRODUCTION

Let G be a finite simple connected graph. Write $d(v)$ for the degree of v, $N(v)$ for its open neighbourhood, and

$$d^*(v) = \frac{1}{d(v)} \sum_{u \in N(v)} d(u), \qquad \delta^*(G) = \min_{v \in V(G)} d^*(v).$$

The distance matrix is $D(G) = (d_G(u, v))_{u,v \in V(G)}$. Its eigenvalues are ordered as

$$\partial_1(G) \geq \cdots \geq \partial_n(G), \qquad \partial_n(G) = \lambda_{\min}(D(G)).$$

Aouchiche and Hansen record the following Graffiti conjecture as Conjecture 7.16 and attribute it to Fajtlowicz's 1998 *Written on the Wall* report [1] (see also [2, Conjecture 7.16]).

Conjecture (WOW-284). *If G has order at least three and girth at least five, then*

$$\delta^*(G) \leq -\lambda_{\min}(D(G)).$$

For graphs in the domain of the conjecture, put

$$\Phi(G) = \delta^*(G) + \lambda_{\min}(D(G)).$$

Thus G is a strict counterexample precisely when $\Phi(G) > 0$.

The initial disproof is short. A degree-k Moore graph of diameter two has

$$A^2 = (k-1)I - A + J, \qquad D = 2J - 2I - A,$$

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and hence

$$\delta^*(G) = k, \qquad \lambda_{\min}(D) = -\frac{3+\sqrt{4k-3}}{2}.$$

The conjecture holds on these graphs exactly for $k \le 3$, with equality at $k = 3$, and fails for every realizable $k > 3$. The degree-seven Hoffman–Singleton graph therefore gives a gap of three.

The purpose of this paper is not merely to list descendants of this graph. It addresses three structural questions.

(1) Which spectral mechanism governs regular counterexamples?
(2) How restrictive are the degree, diameter, and order conditions?
(3) How stable is the counterexample property under deletion?

Our main conclusions are as follows.

- In diameter three, WOW-284 is equivalent to a two-sided adjacency spectrum window centred at -1; see Theorem 3.1.
- Every regular strict counterexample has degree at least six and diameter at most four. A diameter-four example, if one exists, has degree at least ten; see Theorems 4.2, 4.3, and 4.4.
- The standard one-variable nonbacktracking linear-programming hierarchy has exact ceiling

$$B_k = \frac{(k+2)(k^2+3)}{6},$$

  with a unique optimizer up to positive scaling; see Theorem 5.3.
- The LP ceiling gives $n \le 51$ at degree six. An edge-local positive-semidefinite certificate then rules out $n = 51$ by the congruence

$$5N_5 = 153 \cdot 11 = 1683,$$

  so $n \le 50$; see Theorem 6.2.
- One-vertex, adjacent-pair, and nonadjacent-pair deletions of Moore graphs admit exact invariant-subspace decompositions for their recomputed distance matrices; see Section 7.
- Every deletion of at most five vertices from the Hoffman–Singleton graph remains a strict counterexample, and this universal radius is sharp; see Theorem 8.2.

The distance-polynomial viewpoint is established in the theory of minimal cages, distance-polynomial graphs, and spectral excess [3, 4]. Nonbacktracking linear-programming bounds are due to Nozaki and related spectral-Moore work [5, 6]. Our contribution is the specialization to the two-sided WOW window, the exact method ceiling, the edge-local cycle certificate, and the deletion theory developed below. We make no claim that the classical Hoffman–Singleton or Moore spectra are new, that order 38 is minimal, or that the project-derived extensions have uncontested priority. The literature boundaries are returned to in Section 11.

The analytic arguments are proved in the text. Precisely specified finite classifications and matrix certificates are treated as computer-assisted proof components, each with an inline reference to an independent exact verifier. The code is part of release `v2.2.0`; a path such as `scripts/verify_proof_audit_01_edge_local.py` identifies the exact file used for the corresponding audit.

2. Local growth, Moore graphs, and explicit counterexamples

2.1. **Dual degree as radius-two growth.** For $v \in V(G)$, write $\Gamma_i(v)$ for the distance-i sphere and $B_2(v) = \Gamma_0(v) \cup \Gamma_1(v) \cup \Gamma_2(v)$.

Proposition 2.1 (Backelin's second-degree identity)**.** *If G contains no triangle and no 4-cycle, then*

$$|B_2(v)| = 1 + \sum_{u \in N(v)} d(u), \qquad d^*(v) = \frac{|B_2(v)| - 1}{d(v)}.$$

Proof. For distinct neighbours u, w of v, the sets $N(u) \setminus \{v\}$ and $N(w) \setminus \{v\}$ are disjoint; an intersection would form a 4-cycle, and a member in $N(v)$ would form a triangle. These sets partition $\Gamma_2(v)$, so

$$|\Gamma_2(v)| = \sum_{u \in N(v)} (d(u) - 1).$$

Adding the centre and first sphere proves the formula. This is the normalized form of Backelin's second-degree lemma [7, Lemma 2.1]. □

2.2. The Moore threshold.

Theorem 2.2. *Let M be a degree-k Moore graph of diameter two, $k \geq 2$. Then*

$$|V(M)| = k^2 + 1, \qquad g(M) = 5, \qquad \delta^*(M) = k,$$

$$\lambda_{\min}(D(M)) = -\frac{3 + \sqrt{4k-3}}{2},$$

and

$$\Phi(M) = k - \frac{3 + \sqrt{4k-3}}{2}.$$

Thus M satisfies WOW-284 exactly for $k \leq 3$, with equality exactly at $k = 3$.

Proof. The Moore bound is attained, so adjacent vertices have no common neighbour and nonadjacent vertices have exactly one. Therefore

$$A^2 = (k-1)I - A + J.$$

On $\mathbf{1}^{\perp}$, the nonprincipal adjacency eigenvalues are the roots

$$r, s = \frac{-1 \pm \sqrt{4k-3}}{2}.$$

Every nonedge has distance two, hence $D = 2J - 2I - A$. The least distance eigenvalue is $-2 - r = -(3 + \sqrt{4k-3})/2$. Regularity gives $\delta^* = k$, and

$$(2k-3)^2 - (4k-3) = 4(k-1)(k-3)$$

gives the threshold. The exact scalar and finite checks are independently repeated by `scripts/verify_regular_score_calculus.py`. □

2.3. A coordinate Hoffman–Singleton certificate.

All subscripts below lie in $\mathbb{F}_5 = \mathbb{Z}/5\mathbb{Z}$. Let

$$V(M) = \{P_{i,j} : i, j \in \mathbb{F}_5\} \,\dot\cup\, \{Q_{k,\ell} : k, \ell \in \mathbb{F}_5\},$$

with edges

$$\begin{aligned} P_{i,j} &\sim P_{i,j\pm1}, \\ Q_{k,\ell} &\sim Q_{k,\ell\pm2}, \\ P_{i,j} &\sim Q_{k,ik+j}. \end{aligned}$$

This is Hafner's affine-coordinate form of the Hoffman–Singleton graph after a minor reindexing [8].

Proposition 2.3. *The coordinate construction is a simple connected 7-regular graph on 50 vertices. Adjacent pairs have no common neighbour and nonadjacent pairs have exactly one. Consequently it has girth five, diameter two, and*

$$\operatorname{Spec} D(M) = \{91^{(1)}, 1^{(21)}, (-4)^{(28)}\}.$$

In particular, $\delta^*(M) = 7$ and $\Phi(M) = 3$.

Proof. The neighbourhoods are

$$N(P_{i,j}) = \{P_{i,j-1}, P_{i,j+1}\} \cup \{Q_{k,ik+j} : k \in \mathbb{F}_5\},$$
$$N(Q_{k,\ell}) = \{Q_{k,\ell-2}, Q_{k,\ell+2}\} \cup \{P_{i,\ell-ik} : i \in \mathbb{F}_5\}.$$

They have seven distinct entries. For two vertices of the same type, common neighbours are determined by one nonzero linear equation over $\mathbb{F}_5$; for a cross pair $P_{i,j}, Q_{k,\ell}$, put $r = \ell - (ik + j)$. The pair is adjacent for $r = 0$, has one common P-neighbour for $r \in \{\pm 1\}$, and one common Q-neighbour for $r \in \{\pm 2\}$. The five residues are exhausted. The claimed geometry and spectrum now follow from Theorem 2.2. The exhaustive pair certificate, integer BFS distances, characteristic polynomial, and exact positive-definiteness check are in `scripts/verify_exact.py`. ☐

2.4. **Smaller exact counterexamples.** Let

$$\mathcal{P} = \{P_{0,j}, Q_{0,j} : j \in \mathbb{F}_5\}.$$

The induced graph $M[\mathcal{P}]$ is a Petersen graph. Put

$$R = M - \mathcal{P}, \quad H_{39} = R - P_{1,0}, \quad H_{38} = R - \{P_{1,0}, P_{1,1}\},$$

and let X_{42} be the second subconstituent of $P_{0,0}$, namely the graph induced by the vertices at distance two from it.

Theorem 2.4. *The following are strict counterexamples.*

G	$\lvert V(G)\rvert$	$\delta^*(G)$	$\lambda_{\min}(D(G))$
H_{38}	38	17/3	$-3 - \sqrt{7}$
H_{39}	39	35/6	$> -35/6$
R	40	6	-5
X_{42}	42	6	-5
M	50	7	-4

The entry for H_{39} records an exact strict lower bound obtained from positive definiteness of $6D + 35I$; it is not a decimal approximation to the least eigenvalue. Moreover, all 40 *labelled singleton deletions of R, and all* 120 *labelled deletions of the endpoints of an edge of R, are strict counterexamples with the same respective characteristic polynomials.*

Proof. The graph R is the classical $(6, 5)$-cage of O'Keefe and Wong [9, 10]. The Moore block identity gives

$$\operatorname{Spec} A(R) = \{6^{(1)}, 2^{(18)}, 1^{(4)}, (-2)^{(5)}, (-3)^{(12)}\},$$

and Theorem 3.1 below maps this to

$$\operatorname{Spec} D(R) = \{75^{(1)}, 3^{(5)}, 0^{(16)}, (-5)^{(18)}\}.$$

The second-subconstituent calculation gives

$$\operatorname{Spec} D(X_{42}) = \{81^{(1)}, 4^{(6)}, 0^{(14)}, (-5)^{(21)}\}.$$

For H_{38}, a direct degree count gives $\delta^* = 17/3$, while the factor $x^2 + 6x + 2$, together with an exact Sturm isolation, gives the least root $-3 - \sqrt{7}$. For H_{39}, the exact matrix $6D + 35I$ is positive definite. All graph, girth, distance, dual-degree, Sturm, and rational LDL^{T} certificates are checked by `scripts/verify_extended.py`; the labelled deletion families are exhausted by `scripts/verify_descendant_families.py`. No transitivity or numerical root ordering is used. ☐

Theorem 2.5 (Moore second subconstituents)**.** *Let M be a degree-K Moore graph of diameter two, $K \geq 3$, and let X be the graph induced by $\Gamma_2(v)$ for a fixed vertex v. Then X has order $K(K-1)$, degree $K - 1$, girth at least five, diameter three, and*

$$\lambda_{\min}(D(X)) = -\frac{5 + \sqrt{4K - 3}}{2}.$$

Consequently

$$\Phi(X) = K - 1 - \frac{5 + \sqrt{4K - 3}}{2},$$

which is positive exactly for integers $K \geq 6$.

Proof. Relative to $\{v\} \sqcup N(v) \sqcup \Gamma_2(v)$, write the adjacency matrix with incidence block C and induced block B. The Moore identity gives

$$CC^{\mathsf{T}} = (K-1)I, \qquad CB = J - C,$$

and on $\ker C$,

$$B^2 + B - (K-1)I = 0.$$

Thus B has principal eigenvalue $K - 1$, eigenvalue -1 on $C^{\mathsf{T}}(\mathbf{1}^{\perp})$, and the two Moore roots $(-1 \pm \sqrt{4K-3})/2$ on $\ker C$. The remaining vertices sharing the unique neighbour of a chosen vertex are at distance three, so Theorem 3.1 applies. Among the nonprincipal adjacency eigenvalues, $(-1 + \sqrt{4K-3})/2$ maximizes $|\theta + 1|$; substitution gives

$$\lambda_{\min}(D(X)) = -\frac{5 + \sqrt{4K-3}}{2}.$$

The threshold reduces to

$$(2K-7)^2 - (4K-3) = 4(K^2 - 8K + 13) > 0$$

for $K \geq 6$. The finite $K = 7$ instance is checked by `scripts/verify_wow284_38_40_42.py`. □

3. The regular score calculus

Theorem 3.1 (Diameter-three score formula)**.** *Let G be connected, k-regular, of girth at least five and diameter three, with adjacency matrix A and order n. Then*

$$D = 3J + (k-3)I - 2A - A^2,$$

$$D + kI = 3J + (2k-2)I - (A+I)^2.$$

The principal distance eigenvalue is $3n - k^2 - k - 3$, and a nonprincipal adjacency eigenvalue θ gives the distance eigenvalue

$$\mu(\theta) = k - 2 - (\theta + 1)^2.$$

Consequently

$$\boxed{\Phi(G) = 2k - 2 - \max_{\theta \neq k}(\theta + 1)^2.}$$

Thus G is a strict counterexample exactly when

$$|\theta + 1| < \sqrt{2k-2} \qquad (\theta \neq k).$$

Proof. Girth at least five gives the distance-two matrix $A_2 = A^2 - kI$, and diameter three gives $A_3 = J - I - A - A_2$. Substitute in $D = A + 2A_2 + 3A_3$. On $\mathbf{1}^{\perp}$, J vanishes, and regularity gives $\delta^* = k$. The principal distance eigenvalue is positive and is the Perron root because every off-diagonal entry of D is positive. The exact operator and score identities are independently checked by `scripts/verify_regular_score_calculus.py`. □

Corollary 3.2 (Bipartite obstruction)**.** *A connected k-regular bipartite graph of girth at least five and diameter three is not a strict counterexample for $k \geq 3$.*

Proof. The nonprincipal eigenvalue $-k$ violates the open shifted window, with equality only when $k = 3$. □

Proposition 3.3 (Higher-diameter transfer). *Let G be connected and k-regular, with diameter d and girth at least $2d-1$. Define*

$$F_0 = 1, \quad F_1 = x, \quad F_2 = x^2 - k, \quad F_i = xF_{i-1} - (k-1)F_{i-2} \quad (i \geq 3).$$

Then $A_i = F_i(A)$ for $0 \leq i \leq d-1$, and

$$D = dJ + q_d(A), \qquad q_d(x) = \sum_{i=0}^{d-1}(i-d)F_i(x).$$

In particular,

$$q_3(x) = k - 3 - 2x - x^2,$$
$$q_4(x) = -x^3 - 2x^2 + (2k-4)x + 2k - 4.$$

Proof. Up to length $d-1$, the girth condition makes nonbacktracking walks between two vertices unique exactly when their length is the graph distance. Hence the distance matrices are the nonbacktracking polynomials in A; summing $D = \sum iA_i$ and eliminating A_d with $J = \sum A_i$ proves the formula. This lies within the established distance-polynomial framework [3, 4]. □

4. Degree and diameter obstructions

Lemma 4.1. *For every connected graph,*

$$\lambda_{\min}(D(G)) \leq -\operatorname{diam}(G).$$

Proof. For a diametral pair u, v, the Rayleigh quotient of $e_u - e_v$ is $-d_G(u,v)$. □

Theorem 4.2. *Every connected regular strict counterexample to WOW-284 has degree at least six.*

Proof. Let the degree be k. Lemma 4.1 and strictness give $\operatorname{diam}(G) < k$. The cases $k \leq 2$ are immediate. For $k = 3$, the girth lower bound and diameter at most two force the Petersen graph, which is an equality case of Theorem 2.2.

For $k = 4$, diameter two would require a degree-four Moore graph, whose adjacency multiplicities are nonintegral. In diameter three, the exact LP bound of Theorem 5.3 gives $n < 19$, whereas the radius-two ball has 17 vertices and diameter three requires at least one more. Hence $n = 18$. The distance-three matrix is then a perfect matching. Its -1-eigenspace W is a nine-dimensional rational subspace of $\mathbf{1}^{\perp}$, and

$$A_3 = J + 3I - A - A^2 \quad \Longrightarrow \quad (A^2 + A - 4I)|_W = 0.$$

The polynomial $x^2 + x - 4$ is irreducible over $\mathbb{Q}$, so a rational space on which it annihilates an operator has even dimension, a contradiction.

For $k = 5$, diameter two again fails the Moore multiplicity condition. A diametral geodesic in diameter four yields the principal submatrix $D(P_5)$, whose factor $x^2 + 6x + 4$ supplies the eigenvalue $-3 - \sqrt{5} < -5$; Cauchy interlacing excludes this case. In diameter three, Meringer's lower bound and Theorem 5.3 leave $n \in \{30, 31, 32\}$, and parity removes 31. At $n = 32$, the normalized distance-layer compression has nonprincipal factor

$$p_{5,6}(x) = 4x^3 + 10x^2 - 16x - 30, \qquad p_{5,6}(11/6) = -29/27 < 0.$$

Its largest root therefore exceeds $11/6 > -1 + \sqrt{8}$, contradicting the open shifted window. At $n = 30$, Meringer's isomorph-free enumeration leaves exactly four $(5,5)$-cages [11]; each fixed record has an exact distance eigenvalue at most -5. The complete case split, the four fixed graph6 records, characteristic polynomials, and interlacing directions are independently checked by `scripts/verify_proof_audit_04_regular_low_degree.py`. □

Theorem 4.3 (Endpoint-neighbourhood obstruction). *Let G be any connected finite simple graph, and let u, v be vertices at distance $d \geq 5$. Put $p = d(u)$ and $q = d(v)$. Then*

$$\boxed{\lambda_{\min}(D(G)) \leq p + q - 2 - \sqrt{(p-q)^2 + pq(d-2)^2}.}$$

If δ is the ordinary minimum degree, then

$$\boxed{\lambda_{\min}(D(G)) \leq -\delta(d-4) - 2.}$$

Consequently every strict WOW-284 counterexample satisfies

$$\Delta > \delta(d-4) + 2.$$

In particular, every regular strict counterexample has diameter at most four.

Proof. The two endpoint neighbourhoods are disjoint. Give weight $a > 0$ to $N(u)$, weight $-b < 0$ to $N(v)$, and zero elsewhere. Within one neighbourhood distances are at most two; between the two neighbourhoods they are at least $d-2$. Since the cross products are negative, the Rayleigh quotient is at most that of

$$\begin{pmatrix} 2(p-1) & -(d-2)\sqrt{pq} \\ -(d-2)\sqrt{pq} & 2(q-1) \end{pmatrix}.$$

Its least eigenvalue is the first displayed bound, and its least eigenvector can be chosen with both coordinates positive. Write $p = \delta + \alpha$, $q = \delta + \beta$, and $t = d - 2$. The identity

$$(p-q)^2 + pqt^2 - (p + q + \delta(t-2))^2 = (t-2)\{\delta t(\alpha+\beta) + (t+2)\alpha\beta\}$$

is nonnegative and gives the second bound. Finally $\delta^*(G) \leq \Delta$. The sign choice, radical comparison, and integer rounding are independently audited by `scripts/verify_proof_audit_10_endpoint_diameter.py`. □

Theorem 4.4 (Diameter four). *Let G be connected, k-regular, of girth at least five and diameter four. Then*

$$\boxed{\lambda_{\min}(D(G)) \leq -\frac{7 + \sqrt{16k+1}}{2}.}$$

Consequently no such graph of degree $2 \leq k \leq 9$ is a strict counterexample.

Proof. Choose u, v at distance four and put $U = N(u)$, $V = N(v)$. For fixed $a \in U$, distinct vertices $b, b' \in V$ at distance two from a cannot use the same common neighbour, or a 4-cycle results. Thus at most $k(k-1)$ pairs in $U \times V$ have distance two, and

$$\sum_{a \in U, b \in V} d(a,b) \geq 2k^2 + k.$$

Assign weights $\alpha, \beta, -\alpha, -\beta$ to u, U, v, V, respectively. Counting unordered pairs and then doubling gives

$$\frac{x^{\mathsf{T}} D(G) x}{x^{\mathsf{T}} x} \leq \frac{-4\alpha^2 - 4k\alpha\beta - 3k\beta^2}{\alpha^2 + k\beta^2}.$$

After setting $y_1 = \alpha$ and $y_2 = \sqrt{k}\,\beta$, the right-hand side is the Rayleigh quotient of

$$\begin{pmatrix} -4 & -2\sqrt{k} \\ -2\sqrt{k} & -3 \end{pmatrix},$$

whose least eigenvalue is the displayed value and has a positive-coordinate minimizer. The strict comparison with $-k$ holds for $2 \leq k \leq 9$. Every orientation factor, cross-distance sign, and endpoint comparison is independently checked by `scripts/verify_proof_audit_11_diameter_four.py`. □

Corollary 4.5 (Regular trichotomy). *Every regular strict counterexample has one of the following forms:*

(1) diameter two, hence a Moore graph;
(2) diameter three, with all nonprincipal adjacency eigenvalues in the open shifted WOW window;
(3) diameter four, with degree at least ten.

There are no regular strict counterexamples of diameter at least five.

5. Moment bounds and the exact LP ceiling

Let G satisfy the hypotheses of Theorem 3.1, and write its nonprincipal adjacency eigenvalues as θ_i. Put $y_i = \theta_i + 1$.

Proposition 5.1 (Fourth-moment identity). *One has*

$$\sum_{i=1}^{n-1} (2k - 2 - y_i^2)(y_i + 1)^2 = (k+2)\big((k+2)(k^2+3) - 6n\big).$$

Every strict counterexample therefore satisfies

$$\boxed{n < B_k := \frac{(k+2)(k^2+3)}{6}.}$$

Proof. Use

$$\operatorname{tr} A = \operatorname{tr} A^3 = 0, \quad \operatorname{tr} A^2 = nk, \quad \operatorname{tr} A^4 = nk(2k-1),$$

remove the principal eigenvalue, and expand both sides. In a strict counterexample, each factor $2k - 2 - y_i^2$ is positive. The sum cannot vanish: otherwise every nonprincipal adjacency eigenvalue would equal -2, and $\operatorname{tr} A = 0$ would give $k - 2(n-1) = 0$, or $n = (k+2)/2$, incompatible with the elementary bound $n \geq k+1$ for a simple k-regular graph. The identity is checked symbolically by `scripts/verify_degree_six_gate.py` and independently within `scripts/verify_proof_audit_02_two_sided_lp.py`. □

The preceding bound is in fact optimal within the entire standard one-variable nonbacktracking LP hierarchy.

Definition 5.2. Let $F_i = F_i^{(k)}$ be the nonbacktracking polynomials from Proposition 3.3, and set

$$I_k = [-1 - \sqrt{2k-2}, -1 + \sqrt{2k-2}].$$

A finite polynomial $f = \sum_i f_i F_i$ is *admissible* if

$$f_0 > 0, \qquad f_i \geq 0 \quad (i \geq 5), \qquad f(x) \leq 0 \quad (x \in I_k).$$

For a k-regular graph of girth at least five whose nonprincipal spectrum lies in I_k, one has $\operatorname{tr} F_i(A) = 0$ for $1 \leq i \leq 4$, while $\operatorname{tr} F_i(A) \geq 0$ for $i \geq 5$, since these traces count closed nonbacktracking walks. The coefficient conditions therefore give $nf_0 \leq \operatorname{tr} f(A)$. On the other hand, $f(\theta) \leq 0$ for every nonprincipal eigenvalue, so $\operatorname{tr} f(A) \leq f(k)$. Thus

$$nf_0 \leq \operatorname{tr} f(A) \leq f(k), \qquad n \leq \frac{f(k)}{f_0}.$$

Theorem 5.3 (Exact LP ceiling and rigidity). *For every integer $k \geq 4$ and every admissible f,*

$$\boxed{\frac{f(k)}{f_0} \geq B_k = \frac{(k+2)(k^2+3)}{6}.}$$

Equality holds if and only if f is a positive scalar multiple of

$$\boxed{f_*(x) = \frac{(x+2)^2(x^2 + 2x - (2k-3))}{6(k+2)}.}$$

Thus increasing the polynomial degree cannot improve this one-variable LP bound. If the nonprincipal spectrum lies in the interior of I_k, then $n < B_k$.

Proof. The primal expansion is

$$\begin{aligned} 6(k+2)f_*(x) &= 6(k+2)F_0(x) + 2(2k+7)F_1(x) \\ &\quad + (k+13)F_2(x) + 6F_3(x) + F_4(x), \end{aligned}$$

so f_* is admissible and $f_*(k) = B_k$.

For the dual certificate, put $\Delta = \sqrt{2k-2}$, $\xi_\pm = -1 \pm \Delta$, and $\xi_0 = -2$. Define

$$\begin{aligned} w_- &= \frac{k(k+2)(2k^2-6-3(k-1)\Delta)}{24(2k-3)}, \\ w_0 &= \frac{k(k-1)(k^2+3)}{6(2k-3)}, \\ w_+ &= \frac{k(k+2)(2k^2-6+3(k-1)\Delta)}{24(2k-3)}. \end{aligned}$$

All weights are positive. For the only nontrivial one, this follows from

$$(2k^2-6)^2 - 18(k-1)^3 = 2(k-3)(2k-3)(k^2+3) > 0.$$

The measure $\mu = w_-\delta_{\xi_-} + w_0\delta_{\xi_0} + w_+\delta_{\xi_+}$ satisfies

$$\mu(1) = B_k - 1, \qquad \mu(F_i) = -F_i(k) \quad (1 \le i \le 4).$$

For $i \ge 5$, the slack $a_i = \mu(F_i) + F_i(k)$ is strictly positive. For $5 \le i \le 9$, exact calculation gives, after removing the common positive factor $k(k-1)(k+2)(k^2+3)/6$, respectively,

$$2, \quad 5k-13, \quad 2(3k^2-17k+25),$$
$$6k^3-47k^2+139k-150, \quad 2(3k^4-27k^3+106k^2-219k+194),$$

all positive for $k \ge 4$. For $i \ge 10$, put $r = k-1 \ge 3$. The support lies in $[-2\sqrt{r}, 2\sqrt{r}]$, because $1+\sqrt{2r} \le 2\sqrt{r}$, and the Chebyshev representation gives

$$\frac{|\mu(F_i)|}{F_i(k)} \le \frac{2i+1}{3}\,3^{3-i/2} < 1.$$

Since $f \le 0$ on the support of μ, weak duality gives

$$0 \ge \int f\,d\mu \ge B_k f_0 - f(k).$$

If equality holds, strict positivity of every high-degree slack forces $f_i = 0$ for $i \ge 5$. Equality on the positive dual support forces zeros at $\xi_-, -2, \xi_+$; the interior zero -2 has even multiplicity because $f \le 0$ on I_k. Degree at most four then forces f to be a scalar multiple of f_*, and $f_0 > 0$ makes the scalar positive. Equality in the graph bound would then force every nonprincipal adjacency eigenvalue to be -2, which contradicts the trace equation; hence the open-window bound is strict. Every symbolic identity, finite slack, tail bound, and equality-nullspace calculation is independently checked by `scripts/verify_proof_audit_02_two_sided_lp.py`. □

6. Edge-local positivity and the degree-six boundary

Theorem 5.3 gives $n \le 51$ for a degree-six regular strict counterexample. The last order is eliminated by using local cycle data that the one-point LP trace bound does not see.

Proposition 6.1 (Edge-local cycle bounds). *Let G be connected and k-regular, of girth at least five and diameter three, and suppose its nonprincipal adjacency spectrum lies in the closed shifted WOW window. For an edge uv, let σ_{uv} be the number of 5-cycles containing that edge. Then*

$$2k-2 \le \sigma_{uv} \le \frac{2(k+2)^2(k^2+3)}{n} - 10k - 26.$$

If $n = k^2+1+c$, then also

$$\sigma_{uv} \ge (k-1)^2 - c.$$

Proof. Put

$$f_k(x) = (x+2)^2((x+1)^2 - (2k-2)), \quad C_k = f_k(k),$$

and

$$M = -f_k(A) + \frac{C_k}{n} J.$$

The spectral window gives $M \succeq 0$. On an edge,

$$(A^3)_{uv} = 2k - 1, \qquad (A^4)_{uv} = \sigma_{uv},$$

so the diagonal and edge entries of M are

$$a = \frac{C_k}{n} - 6(k+2), \quad b = \frac{C_k}{n} - (4k+14) - \sigma_{uv}.$$

The 2×2 principal minor gives $-a \le b \le a$, yielding the first two bounds with their stated directions.

For the last bound, radius-two balls have size $k^2 + 1$, and

$$|B_2(u) \cap B_2(v)| = 2k + \sigma_{uv}.$$

The vertices at distance two from both endpoints are in bijection with the 5-cycles through uv. Inclusion–exclusion gives the result. The walk classification, sign directions, and radius-two bijection are independently checked by `scripts/verify_proof_audit_01_edge_local.py`. □

Theorem 6.2. *Every connected* 6*-regular strict counterexample to WOW-284 has order at most* 50*.*

Proof. A separate Rayleigh and trace argument shows that any degree-six strict counterexample has diameter three. For vertices at distance $d \ge 4$, the vector with weights $3, 1, -3, -1$ on the two endpoints and their respective neighbourhoods has Rayleigh quotient at most

$$\frac{204 - 81d}{15} \le -8,$$

contradicting $\delta^* = 6$. Diameter two would force $(x-6)(x^2+x-5)^{18}$; its root sum is -12, contradicting $\operatorname{tr} A = 0$. The LP ceiling gives $n \le 51$.

Assume $n = 51$, so $c = 14$. Proposition 6.1 gives, for every edge,

$$11 \le \sigma_{uv} \le \frac{202}{17} < 12.$$

Hence $\sigma_{uv} = 11$. The graph has 153 edges, and counting incidences between edges and 5-cycles gives

$$5N_5 = 153 \cdot 11 = 1683,$$

impossible modulo five. The full diameter reduction and incidence audit are contained in `scripts/verify_proof_audit_01_edge_local.py`. □

Corollary 6.3 (Low-degree order windows)**.** *The current bounds imply:*

$$\begin{aligned} k = 6 : &\quad n \le 50, \\ k = 7 : &\quad n = 50 \textit{ in diameter two, or } n \le 76 \textit{ in diameter three}, \\ k = 8 : &\quad n \le 110, \\ k = 9 : &\quad n \le 152. \end{aligned}$$

Proof. Apply Theorems 4.4 and 5.3, then use the parity of kn. At $k = 8, n = 111$, the edge-local bounds force every edge into exactly 14 five-cycles, but

$$5N_5 = 14 \cdot 444 = 6216$$

is impossible. Exact integer arithmetic is checked by `scripts/verify_low_degree_windows.py`. □

6.1. **Necessary structure at order fifty.** The remaining degree-six boundary is highly constrained, although not yet eliminated.

Theorem 6.4. *Let G be a connected 6-regular graph of order 50 and girth at least five, and suppose $\Phi(G) > 0$. Then G has diameter three. Every edge lies in 12 or 13 five-cycles. Let H be the spanning subgraph of edges of the second type, let $m = |E(H)|$, and let $\tau(v)$ be the number of five-cycles through v. Then*

$$\tau(v) \in \{36, 37, 38\}, \qquad d_H(v) = 2\tau(v) - 72 \in \{0, 2, 4\},$$
$$m \equiv 0 \pmod 5, \qquad N_5 = 360 + \frac{m}{5}.$$

For a two-edge path $u - v - w$, put

$$r_{uvw} = 6\alpha_{uvw} + \beta_{uvw},$$

where α and β count the five- and six-cycles containing the path. The allowed values are

types of the two incident edges	r_{uvw}
low–low	$30, 31, 32$
mixed	$30, 31, 32$
high–high	$30, 31.$

Writing $S_2 = \sum_v d_H(v)^2$, one has

$$1950 - m \le N_6 \le 2200 - \frac{5m}{6} - \frac{S_2}{12},$$
$$\frac{43m^2 - 70200m + 119632500}{58500} \le N_6 \le \frac{4220000 - 2200m - 7m^2}{2000}.$$

Exact enumeration of the resulting coarse degree profiles leaves 266 possibilities.

Proof. By Theorem 6.2, the diameter is three. Around a fixed vertex the distance layers have sizes $1, 6, 30, 13$. If τ is the number of five-cycles through the centre, the average row quotient is similar to the symmetric compression on normalized layer indicators. Its nonprincipal factor q_τ satisfies

$$195q_\tau(-1 + \sqrt{10}) = (-215 + 56\sqrt{10})\tau + 7350 - 1860\sqrt{10},$$

and $195q_\tau(6) = 1500(75 - \tau) > 0$. Hence $\tau \ge 39$ would create a compression eigenvalue above the open WOW endpoint. Conversely, the number $150 - 2\tau$ of edges from the second to the third layer is at most $13 \cdot 6$, so $\tau \ge 36$. Thus $\tau \in \{36, 37, 38\}$, and edge–cycle incidence gives the even high-edge subgraph and the formula for N_5.

For a two-path $u - v - w$, girth at least five gives $(A^3)_{uw} = \alpha_{uvw}$ and $(A^4)_{uw} = 16 + \beta_{uvw}$. The corresponding 3×3 principal minor of the centered positive-semidefinite matrix yields the displayed finite sets for r_{uvw}, except for an apparent equality value $r = 29$. At equality, the Gram norm of $e_u - e_w$ vanishes, so this vector lies in the kernel of the centered matrix. On $\mathbf{1}^\perp$, that kernel is the adjacency -2-eigenspace; however, the u-coordinate of $A(e_u - e_w)$ is zero because $u \not\sim w$, whereas the u-coordinate of $-2(e_u - e_w)$ is -2. This excludes $r = 29$. Summing the local inequalities gives the first pair of N_6 bounds; shifted moment and localizing matrices give the second. The complete symbolic determinants, Schur complements, kernel argument, and integer enumeration are independently checked by `scripts/verify_proof_audit_06_order50_feasibility.py`. The surviving 266 profiles are exact pruning data, not an existence or nonexistence claim. $\square$

7. Distance spectra of punctured Moore graphs

Let M be a degree-k Moore graph of diameter two and put $\Delta = \sqrt{4k - 3}$.

Theorem 7.1 (One deleted vertex)**.** *For $H = M - v$,*

$$|V(H)| = k^2, \qquad \delta^*(H) = k - \frac{1}{k}, \qquad \lambda_{\min}(D(H)) = -2 - \sqrt{k}.$$

The complete distance spectrum is

$$\operatorname{Spec} D(H) = \{\rho_+, \rho_-\} \cup \{(-2+\sqrt{k})^{(k-1)}, (-2-\sqrt{k})^{(k-1)}\} \\ \cup \left\{ \left(-\frac{\Delta+3}{2}\right)^{(m_+)}, \left(\frac{\Delta-3}{2}\right)^{(m_-)} \right\},$$

where

$$\rho_\pm = k^2 - 2 \pm \sqrt{k(k^3 - 2k^2 + 3k - 1)},$$
$$m_\pm = \frac{k(k-2)(\Delta \pm 1)}{2\Delta}.$$

Thus

$$\Phi(H) = k - \frac{1}{k} - 2 - \sqrt{k},$$

which is positive exactly for integers $k \geq 5$.

Theorem 7.2 (Endpoints of an edge)**.** *Let $uv \in E(M)$ and $H = M - \{u, v\}$. For $k \geq 3$,*

$$|V(H)| = k^2 - 1, \qquad \delta^*(H) = k - \frac{2}{k}, \qquad \lambda_{\min}(D(H)) = -2 - \sqrt{k}.$$

The complete distance spectrum is

$$\operatorname{Spec} D(H) = \{\sigma_+, \sigma_-, k-4\} \\ \cup \{(-2+\sqrt{k})^{(2k-4)}, (-2-\sqrt{k})^{(2k-4)}\} \\ \cup \left\{ \left(-\frac{\Delta+3}{2}\right)^{(a_+)}, \left(\frac{\Delta-3}{2}\right)^{(a_-)} \right\},$$

where

$$\sigma_\pm = k^2 - 3 \pm \sqrt{k^4 - 2k^3 + 3k^2 - 8k + 7},$$
$$a_+ = \frac{(k-2)(k+(k-2)\Delta)}{2\Delta}, \quad a_- = \frac{(k-2)((k-2)\Delta - k)}{2\Delta}.$$

Thus

$$\Phi(H) = k - \frac{2}{k} - 2 - \sqrt{k},$$

which is positive exactly for integers $k \geq 5$.

Proof architecture for Theorems 7.1 and 7.2. For one deleted vertex, put $A = N(v)$, $B = \Gamma_2(v)$, let C be the A-by-B incidence matrix, and let B_0 be the adjacency matrix on B. Recomputed distances give

$$D(H) = \begin{pmatrix} 3(J-I) & 2J - C \\ 2J - C^\top & 2(J-I) - B_0 \end{pmatrix}.$$

The identities $CC^\top = (k-1)I$, $CB_0 = J - C$, and the bottom-right block of the Moore relation yield the orthogonal decomposition

$$\mathbb{R}^{V(H)} = W_{\text{const}} \perp W_{\text{inc}} \perp \ker C, \qquad 2 + 2(k-1) + k(k-2) = k^2.$$

On each zero-sum incidence module the action matrix is $\left(\begin{smallmatrix} -3 & -(k-1) \\ -1 & -1 \end{smallmatrix}\right)$, giving $-2 \pm \sqrt{k}$; on $\ker C$, $B_0^2 + B_0 - (k-1)I = 0$ and $D = -2I - B_0$. The normalized constant quotient supplies $\rho_\pm$.

For an adjacent deleted pair, the cells $A = N(u) \setminus \{v\}$, $B = N(v) \setminus \{u\}$, and the residual cell give an antisymmetric constant line, a symmetric two-dimensional quotient, two zero-sum incidence modules, and a residual kernel. Their dimensions add to

$$3 + 2(2k-4) + (k-2)^2 = k^2 - 1.$$

The same incidence action gives $-2 \pm \sqrt{k}$, the residual Moore relation gives the $a_\pm$ factors, and positive definiteness of the two shifted constant quotients shows that no quotient or residual root lies below $-2 - \sqrt{k}$. Exact quotient normalization, trace-to-multiplicity equations, replacement paths, and all least-root comparisons are independently checked by `scripts/verify_proof_audit_05_small_moore_punctures.py`. □

Theorem 7.3 (Two nonadjacent deleted vertices)**.** *Let u, v be nonadjacent vertices of M, $k \geq 5$, and put $H = M - \{u, v\}$. Define*

$$\begin{aligned} R_k(x) = x^4 &+ (10 - 2k^2)x^3 + (2k^3 - 17k^2 - 2k + 36)x^2 \\ &+ (12k^3 - 49k^2 - 4k + 53)x \\ &- 2k^4 + 17k^3 - 38k^2 + 5k + 20, \end{aligned}$$

$$M_- = \frac{k(k-2) + (k^2 - 4k + 2)\Delta}{2\Delta}, \quad M_+ = \frac{-k(k-2) + (k^2 - 4k + 2)\Delta}{2\Delta}.$$

Then

$$\begin{aligned} \chi_{D(H)}(x) = (x - k + 3)&R_k(x)(x^2 + 4x - k + 3)^{k-2} \\ &\cdot (x^2 + 4x - k + 5)^{k-2} \left(x + \frac{\Delta + 3}{2}\right)^{M_-} \\ &\cdot \left(x - \frac{\Delta - 3}{2}\right)^{M_+}. \end{aligned}$$

Moreover,

$$\delta^*(H) = k - \frac{2}{k},$$

and H is a strict counterexample for every realizable integer $k \geq 6$.

Proof. The deleted vertices have a unique common neighbour w. With $A = N(u) \setminus \{w\}$, $B = N(v) \setminus \{w\}$, $C = N(w) \setminus \{u, v\}$, and residual cell Z, the surviving graph has an equitable five-cell geometry and a perfect matching between A and B. Explicit length-three paths show that every pair whose unique length-two path was deleted has new distance exactly three. The five-cell distance quotient has characteristic polynomial $(x - k + 3)R_k(x)$.

Let R_A, R_B, R_C be the incidence matrices from the three nonconstant cells to Z, and let T be the adjacency matrix on Z. The Moore identity splits the remaining space into matched symmetric and antisymmetric modules, a common-neighbour module, and $K = \ker R_A \cap \ker R_B \cap \ker R_C$. The first three modules give the two displayed quadratic factors and $k - 3$ copies of each Moore linear factor. On K,

$$T^2 + T - (k-1)I = 0, \qquad D = -2I - T.$$

The constant direction of Z has T-eigenvalue $k - 3$, the symmetric, antisymmetric, and common-neighbour images have eigenvalues $-2, 0, -1$ with dimensions $k - 2, k - 2, k - 3$. Since T has zero diagonal, its total trace is zero, and therefore $\operatorname{tr}(T|_K) = 2(k - 2)$. Dimension and trace now give the residual multiplicities; after adding the common-neighbour copies they are precisely M_- and M_+. The full dimension count is

$$5 + 4(k-2) + 2(k-3) + (k-2)(k-4) = k^2 - 1.$$

For strictness, the distance-increase matrix is the adjacency matrix of two copies of K_k meeting in w, and hence has least eigenvalue

$$\lambda_{\min}(E_S) = \frac{k - 2 - \sqrt{k^2 + 4k - 4}}{2}.$$

Proposition 7.4 gives a positive lower bound for the child score for every $k \geq 6$. After the sign-preserving squarings, the remaining condition is $P(k) > 0$, where

$$P(k) = 4k^8 - 39k^7 + 95k^6 + 9k^5 - 173k^4 - 36k^3 + 116k^2 + 80k + 16.$$

Writing $k = m + 6$ makes every coefficient positive. The direct sum, injectivity, orthogonality, all recomputed distances, and the polynomial sign are independently checked by `scripts/verify_proof_audit_03_nonadjacent_puncture.py` and `scripts/verify_research_extensions_exact.py`. □

Proposition 7.4 (Deletion stability)**.** *Let G be connected, $S \subseteq V(G)$, and suppose $H = G - S$ is connected. Put*

$$D_0 = D(G)[V(H)], \qquad E_S = D(H) - D_0,$$
$$a = \delta^*(G), \quad b = \delta^*(H), \quad \gamma = \Phi(G).$$

Then

$$\boxed{\Phi(H) \geq \gamma - (a-b) + \lambda_{\min}(E_S).}$$

Proof. One has

$$D(H) + bI = (D_0 + aI) + E_S - (a-b)I.$$

Principal-submatrix interlacing gives $\lambda_{\min}(D_0 + aI) \geq \gamma$, and Weyl's inequality gives the result. For Moore punctures, E_S is an explicit distance-increase graph; the exact specializations are checked by `scripts/verify_research_extensions_exact.py`. □

8. Small punctures and exact Hoffman–Singleton robustness

Theorem 8.1 (Small-puncture normal form)**.** *Let M be a degree-k Moore graph, let $S \subseteq V(M)$ have size $s \leq k-1$, and put $H = M - S$. Then H is connected, has diameter at most three, and*

$$\boxed{\delta^*(H) = k - \frac{s}{k}.}$$

Let B be the surviving-vertex by deleted-vertex incidence matrix. Then

$$\boxed{D(H) = 2(J-I) - A(H) + BB^{\mathsf{T}} - \operatorname{diag}(BB^{\mathsf{T}}).}$$

Proof. If the unique common neighbour of two surviving nonadjacent vertices is deleted, the Moore geometry constructs $k-1$ internally vertex-disjoint length-three replacement paths. Since at most $s-1 \leq k-2$ further vertices are deleted, one survives. Thus the distance increases from two to exactly three precisely when a deleted vertex was the unique common neighbour, which proves the matrix formula.

For $x \in V(H)$, let $t_x = |N_M(x) \cap S|$. Then

$$\sum_{y \in N_H(x)} t_y \leq s - t_x$$

and

$$d_H^*(x) \geq k - \frac{s - t_x}{k - t_x} \geq k - \frac{s}{k}.$$

The intersection bound

$$\left| \bigcap_{z \in S} \Gamma_2(z) \right| \geq k^2 + 1 - s(k+1) \geq 2$$

provides a surviving vertex x at distance two from every deleted vertex. For $z \in S$, let y_z be the unique common neighbour of x, z. If $y_z \in S$, then the choice of x would require $d_M(x, y_z) = 2$, contradicting $x \sim y_z$. Thus all witnesses survive, each deleted vertex contributes exactly once to the neighbour-degree deficit, and equality is attained. The replacement paths, boundary case $s = k-1$, distance formula, and attainment step are independently checked by `scripts/verify_proof_audit_12_small_puncture.py`. □

Theorem 8.2 (Hoffman–Singleton robustness radius)**.** *Let M be the Hoffman–Singleton graph. Every induced graph $M - S$ with $|S| \leq 5$ is a strict counterexample to WOW-284. This is sharp in the universal sense: there exists a six-vertex set whose deletion is not strict. Hence the universal vertex-deletion robustness radius is exactly five.*

Proof. Theorem 8.1 gives
$$\delta^*(M-S) = \frac{49-|S|}{7}.$$
Two explicitly stored permutations are verified edge-by-edge as automorphisms of the coordinate graph; the group they generate has order 252000. Their orbits on deletion sets of sizes $0, 1, 2, 3, 4, 5$ have counts
$$1, 1, 2, 4, 11, 33.$$
For every one of the 52 representatives, exact fraction arithmetic gives
$$7D(M-S) + (49-|S|)I \succ 0.$$
Orbit-size sums equal $\binom{50}{s}$, so every labelled set is covered.

For sharpness, delete
$$\{P_{2,4}, P_{3,1}, P_{3,4}, Q_{2,1}, Q_{3,4}, Q_{4,4}\}.$$
The resulting graph has $\delta^* = 43/7$, while an exact LDL^T decomposition of $7D + 43I$ has exactly one negative and no zero pivot. Thus it is not strict. Generator action, orbit exhaustion, BFS distances, the small-puncture formula, and a handwritten rational LDL^T implementation are independently checked by `scripts/verify_proof_audit_13_hs_robustness.py`. □

Remark 8.3. The theorem asserts that every deletion through size five succeeds and that at least one deletion of size six fails. It does not assert that every six-vertex deletion fails.

9. Equality and obstructions to natural construction families

Theorem 9.1 (Equality boundary). *Let G be connected, k-regular, of girth at least five and diameter three. Then*
$$\Phi(G) = 0 \iff \max_{\theta \neq k} |\theta + 1| = \sqrt{2k-2}.$$
Equivalently, $D + kI$ is positive semidefinite and singular. If $2k-2$ is not a square, the two boundary adjacency eigenvalues occur with equal multiplicity, so the distance eigenvalue $-k$ has even multiplicity. If $2k-2$ is a square, then $k = 2r^2 + 1$ for some integer r.

Proof. This is immediate from Theorem 3.1. In the nonsquare case, the two boundary values are algebraic conjugates, so their multiplicities in the integral characteristic polynomial agree. In the square case, $2k-2 = s^2$ forces s even. The exact scalar audit is `scripts/verify_equality_boundary.py`. □

Jørgensen's 9-regular order-96 graph of girth five is an exact equality case:
$$\delta^* = 9, \qquad \lambda_{\min}(D) = -9,$$
with multiplicity eight. The construction is due to Jørgensen [12]; the three local graph representations, handwritten graph6 decoder, characteristic polynomials, root intervals, and provenance boundary are checked by `scripts/verify_proof_audit_09_jorgensen96.py`.

9.1. A prime-field obstruction.

For an odd prime $q \geq 7$ and $1 \leq m \leq q$, define $G(q, m)$ on vertices $P_{i,j}, Q_{k,\ell}$, with $0 \leq i, k < m$ and $j, \ell \in \mathbb{F}_q$, by
$$P_{i,j} \sim P_{i,j\pm1},$$
$$Q_{k,\ell} \sim Q_{k,\ell\pm2},$$
$$P_{i,j} \sim Q_{k,ik+j}.$$
This is a balanced specialization of known finite-field girth-five constructions [13]. It is $(m+2)$-regular. A coordinate common-neighbour calculation shows that it has no triangle or 4-cycle for $q \geq 7$: for a cross pair, the possible same-side common neighbours require residues in the disjoint sets $\{\pm1\}$ and $\{\pm2\}$.

Theorem 9.2. *If $G(q,m)$ has diameter three, then it is not a strict counterexample to WOW-284.*

Proof. The zero Fourier block has eigenvalues $m+2$, $2-m$, and 2 with multiplicity $2m-2$; the shifted WOW window leaves only $m \in \{4,5,6\}$. Let $\omega = e^{2\pi i/q}$. On the nonzero character $t=1$, the adjacency block has form

$$\begin{pmatrix} aI & M \\ M^* & bI \end{pmatrix}, \qquad M_{ik} = \omega^{ik},$$

and $\|M\|_F^2 = m^2$, so its largest singular value is at least $\sqrt{m}$. The block therefore has a nonprincipal eigenvalue at least

$$\sqrt{m} + \cos(\pi/7) - \frac{1}{2},$$

which lies above $-1+\sqrt{2m+2}$ for $m = 4,5,6$. The common-neighbour case split, Fourier reduction, radical comparisons, and exact $q = 7$ controls are independently checked by `scripts/verify_proof_audit_08_prime_field.py`. □

9.2. **Layer-respecting matching deletions.** For $\pi \in S_5$, delete the perfect matching

$$\mathcal{M}_\pi = \{P_{i,j}Q_{\pi(i),i\pi(i)+j} : i,j \in \mathbb{F}_5\}$$

from the Hoffman–Singleton graph.

Theorem 9.3. *Each deletion produces a connected simple* 6*-regular graph of order* 50*, girth five, and diameter four. The* 120 *labelled graphs form exactly two isomorphism classes. The* 20 *affine permutations have*

$$\lambda_{\min}(D) = -13, \qquad \Phi = -7,$$

and the 100 *nonaffine permutations have*

$$\lambda_{\min}(D) = -6-\sqrt{61}, \qquad \Phi = -\sqrt{61}.$$

Thus every member strictly satisfies the conjectured inequality.

Proof. Every $P_{i,j}$ occurs once in $\mathcal{M}_\pi$; for a fixed $Q_{k,\ell}$, the unique incident matching edge is obtained from $i = \pi^{-1}(k)$ and $j = \ell - ik$. Thus $\mathcal{M}_\pi$ is a perfect matching, and its deletion leaves a simple 6-regular graph. Deleting edges cannot create a short cycle, while the same-layer pentagons remain; exact breadth-first search gives connectedness and diameter four.

Explicit type-preserving and type-swapping coordinate automorphisms generate orbits of sizes 20 and 100. The representatives have different adjacency characteristic polynomials, so the orbits are distinct isomorphism classes. Exact distance characteristic polynomials and Sturm separators give the stated least roots. All 120 matchings, 400 coordinate maps, 48000 matching images, orbit coverage, graph hypotheses, and root certificates are checked by `scripts/verify_proof_audit_07_layer_matchings.py`. □

10. Exact computation and formal verification

The analytic arguments above are primary. Exact computation is used in three roles: to certify explicitly labelled finite graphs, to exhaust precisely specified finite orbit families, and to check symbolic identities whose written derivations are also supplied. No theorem-level sign or eigenvalue ordering uses floating-point arithmetic.

The principal independent audit entry points are:

Result	Exact verifier
Explicit orders $38, 39, 40, 42, 50$	`scripts/verify_extended.py`
Regular degree at least six	`scripts/verify_proof_audit_04_regular_low_degree.py`
Endpoint diameter theorem	`scripts/verify_proof_audit_10_endpoint_diameter.py`
Diameter-four theorem	`scripts/verify_proof_audit_11_diameter_four.py`
LP ceiling and rigidity	`scripts/verify_proof_audit_02_two_sided_lp.py`
Degree-six order bound	`scripts/verify_proof_audit_01_edge_local.py`
Order-50 feasibility	`scripts/verify_proof_audit_06_order50_feasibility.py`
One-vertex and edge punctures	`scripts/verify_proof_audit_05_small_moore_punctures.py`
Nonadjacent puncture	`scripts/verify_proof_audit_03_nonadjacent_puncture.py`
Small-puncture normal form	`scripts/verify_proof_audit_12_small_puncture.py`
Hoffman–Singleton radius	`scripts/verify_proof_audit_13_hs_robustness.py`
Equality graph	`scripts/verify_proof_audit_09_jorgensen96.py`
Prime-field obstruction	`scripts/verify_proof_audit_08_prime_field.py`
Matching deletion classes	`scripts/verify_proof_audit_07_layer_matchings.py`

The explicit 50-vertex counterexample is fully formalized in Lean 4.31 with Mathlib 4.31 [14, 15]. The development checks the coordinate graph, regularity, the complete common-neighbour certificate, girth, the adjacency-square and distance-matrix identities, and an exact rational diagonalization with multiplicities. Lean also kernel-checks finite spectral certificates for the orders $38, 39, 40, 42$. For orders $38, 39, 42$, these are positive-definiteness certificates for the corresponding shifted distance matrices; the order-40 endpoint contains a complete exact diagonalization. The non-50 formal statements are deliberately described as finite spectral certificates rather than as a single Mathlib theorem bundling every graph-theoretic phrase. The LP ceiling, optimizer rigidity, punctured-Moore spectra, and deletion-robustness theorems remain conventional analytic proofs with exact executable audits; they are not included in the Lean claim above.

The release audit rejects `sorry`, `admit`, `native_decide`, `bv_decide`, unsafe declarations, and new axioms. Representative axiom reports contain only `propext`, `Classical.choice`, and `Quot.sound`.

11. Scope, literature boundary, and open questions

Several ingredients belong to established theory and should not be conflated with the project-derived statements.

- Moore graphs, the Hoffman–Singleton graph, and their intact adjacency and distance spectra are classical [3, 16].
- Proposition 2.1 is Backelin's second-degree identity in normalized form [7].
- Proposition 3.3 belongs to the established distance-polynomial framework [4].
- The one-variable nonbacktracking LP framework is due to Nozaki [5]; Theorem 5.3 is the exact two-sided WOW specialization and optimizer theorem derived here.

- The four $(5,5)$-cages used in Theorem 4.2 come from Meringer's published exhaustive generation [11]; the repository independently verifies the four fixed records but does not replace the enumeration.
- Literature on vertex-deleted adjacency spectra and subconstituents is close in language but does not directly give the recomputed ordinary distance spectra in Section 7 [17, 18].

Targeted searches, recorded in `research-notes/LITERATURE_AUDIT_EXTENSIONS.md` and `research-notes/NEXT_DIRECTION_SOURCE_MATRIX.json`, did not locate direct prior statements of the punctured-Moore distance spectra, the exact two-sided LP optimizer, the edge-local WOW cycle inequality, or the endpoint-neighbourhood bound. Search silence is not proof of novelty. Accordingly, we use "derive" and "prove" rather than "first" or "previously unknown", and make no minimum-order claim.

The main open questions are:

(1) Is there a counterexample of order below 38, and what is the true minimum order?
(2) Does a 6-regular order-50 strict counterexample exist?
(3) Can the one-point and three-vertex constraints of Theorem 6.4 be completed by a multipoint semidefinite or canonical-generation argument?
(4) Do regular diameter-four counterexamples exist for degree at least ten?
(5) Is there an unconditional infinite family of strict counterexamples? Any infinite regular family must have unbounded degree, since Corollary 4.5 and the Moore bound leave finitely many graphs at each fixed degree.
(6) Which deleted sets in a general Moore graph preserve strictness, beyond the metric normal form of Theorem 8.1?

OpenAI ChatGPT-5.6 Sol Pro assisted with adversarial proof checking, literature search, exact verification code, and Lean formalization. No AI system is an author. The author assumes full responsibility for the mathematics, attribution, and conclusions. The source, exact certificates, and build instructions are available at `github.com/SamPetkov/wow284` and correspond to release `v2.2.0`.

Appendix A. Characteristic polynomials for the smallest explicit descendants

For every labelled singleton deletion of R,

$$\begin{aligned} P_{39}(x) = x^9(x+5)^{12}(x^2+6x+3) \\ \cdot (x^3+3x^2-15x-7)^2(x^3+3x^2-15x-3)^2 \\ \cdot (x^4-78x^3+303x^2-70x-450). \end{aligned}$$

For every labelled edge-endpoint deletion of R,

$$\begin{aligned} P_{38}(x) = x^4(x-2)(x+5)^8(x^2+6x+2)^2 \\ \cdot (x^3+3x^2-15x-3) \\ \cdot (x^4+5x^3-7x^2-23x-6) \\ \cdot (x^5+9x^4+7x^3-77x^2-54x-4) \\ \cdot (x^9-67x^8-404x^7+1772x^6+7205x^5 \\ -18489x^4-17018x^3+20288x^2+16824x+1680). \end{aligned}$$

The fixed graph6, adjacency-list, edge-list, and distance-matrix records are in `data/graphs/`.

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